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Lucia Faravelli, Editor
Structural Control and Health Monitoring
Wiley

Dear Editor,

Submission of an original research article: “Blind loci and design criteria for integral load-monitoring channels on continuous girder bridges, with laboratory verification and an assessment on public field records”

We submit the above manuscript for consideration as a Research Article in *Structural Control and Health Monitoring*.

Influence-area bridge weigh-in-motion, introduced by Ojio and Yamada, estimates the gross weight of a vehicle from the area under one response record. That this area equals the response to a uniform unit load is classical, and so is the combination of several channels by generalised least squares. The manuscript states both as such and uses them to derive design criteria for continuous girders, on which positive and negative influence areas partly cancel.

We claim as new:

1. a closed-form catalogue of blind loci for deflection, strain, rotation, shear and reaction channels on two- and three-span and haunched girders, including a near-blind ring for strain gauges set back from the piers;
2. a strain indicator that needs no reference stiffness, a cancellation ratio that separates cancellation from restraint, and a corrected normalisation of the deflection indicator;
3. a placement rule: optimal sensors lie at the extremes of the uniform-load response, for strain at the extremes of $c\,M_q/E\,I$;
4. an identifiability analysis under uncertain pier stiffness, in which the constant follows a Möbius law in one stiffness ratio, and which names the measurement that resolves it;
5. an evidence-based assessment, on a laboratory beam and on public moving-truck records from a five-span box girder, of what single- and multi-channel integral estimators can and cannot deliver.

The field assessment reports every channel with 95 % confidence intervals and equivalence tests, and it reports negative results. The reference weights on these records vary so little that predicting the mean weight beats every estimator, so the records cannot test weight discrimination. Constants are consistent with axle-configuration invariance but the test is inconclusive at better than about 27 %. Generalised least squares with the measured noise covariance did worse out of sample than the plain gauge average, because errors during crossings are much less correlated between gauges than vehicle-free noise; we present this as a design caution for offset-immune estimation. An application to an in-service 55 + 90 + 55 m box girder shows that its side-span deflection gauge is not identifiable from static acceptance data, and predicts the measurements that would decide.

We believe the paper fits the journal's interest in sensor placement, identifiability, signal processing and field measurement. The journal has recently published work on deflection influence lines of continuous beam bridges, which we cite and distinguish.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 licence). The laboratory data, the derived data behind every figure and table, and all analysis scripts with their console output are provided as supplementary material, and the processed in-service bridge data will be deposited at [NEED VALUE: repository name and DOI].

The manuscript is original, has not been published, and is not under consideration by any other journal. [NEED VALUE: two related papers by the authors are under review at *Structures*; state here whether they share the laboratory rig, data or text with this manuscript and how they are cited, or confirm that there is no overlap, and attach them as papers for the editor.] All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Yours sincerely,

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